

Simultaneous Tracking and Segmentation of Carotid Plaques

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Motivation and Objectives

- Stroke is a leading cause of death worldwide. The state of the carotid plaque indicates the likelihood of a stroke occurring. No mature segmentation network currently exists to help evaluate the plaque.
- Ultrasound is a widespread and effective way of imaging the carotid artery because it is low-cost and noninvasive. However, using ultrasound imaging can present issues such as lighting, fuzziness, and occlusion.
- We want to create a segmentation network that can help diagnose the plaque inside the carotid artery by segmenting the carotid plaque in the ultrasound images. Hereby, we want to maximize the accuracy to keep the chance of a false diagnosis minimal.

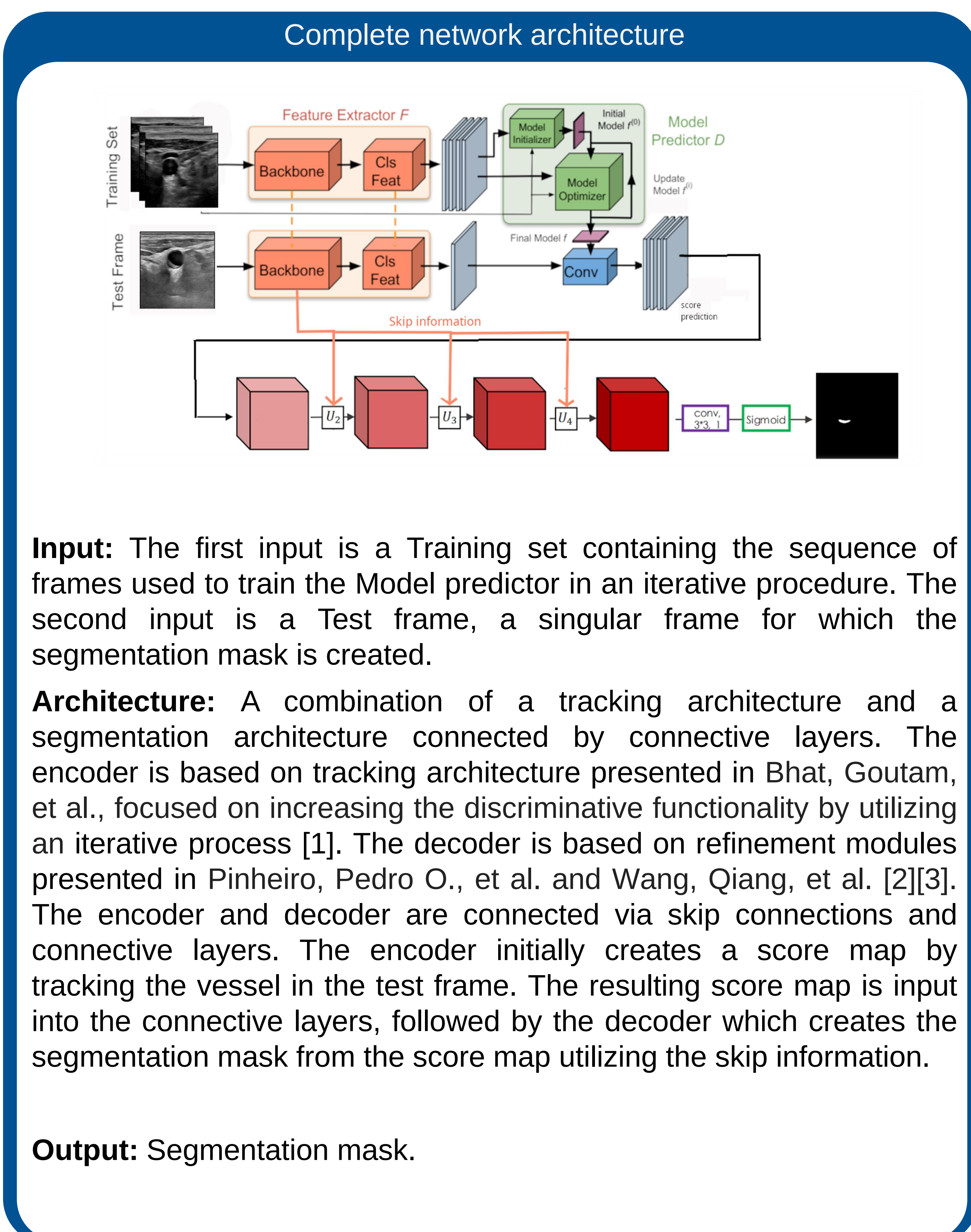
State of the Art and Positioning

Bhat, Goutam, et al. presented a tracking network architecture that used a meta-learning-like iterative process to maximize the discriminative functionality [1]. We used this architecture as the base for our encoder.

Wang, Qiang, et al. proposed a Siamese network architecture with a tree branch decoder in which the bounding box, as well as the segmentation mask, were created [3]. The segmentation branch used refinement modules as a base [2]. We used the segmentation branch as the base for our decoder.

The state-of-the-art network architecture we compared our network performance to is the DeepLab3+ proposed by Chen, Liang-Chieh, et al. [4]. The Deeplab3+ architecture combined the DeepLab3, which used atrous convolutions, with a simple decoder.

Method



Experiments and Data Sets

The dataset was created in collaboration with Klinikum West. We received the B-Mode ultrasound image sequences of patients' carotid ultrasounds and manually segmented the carotid plaque and vessel to create the ground truth masks and bounding box annotations.

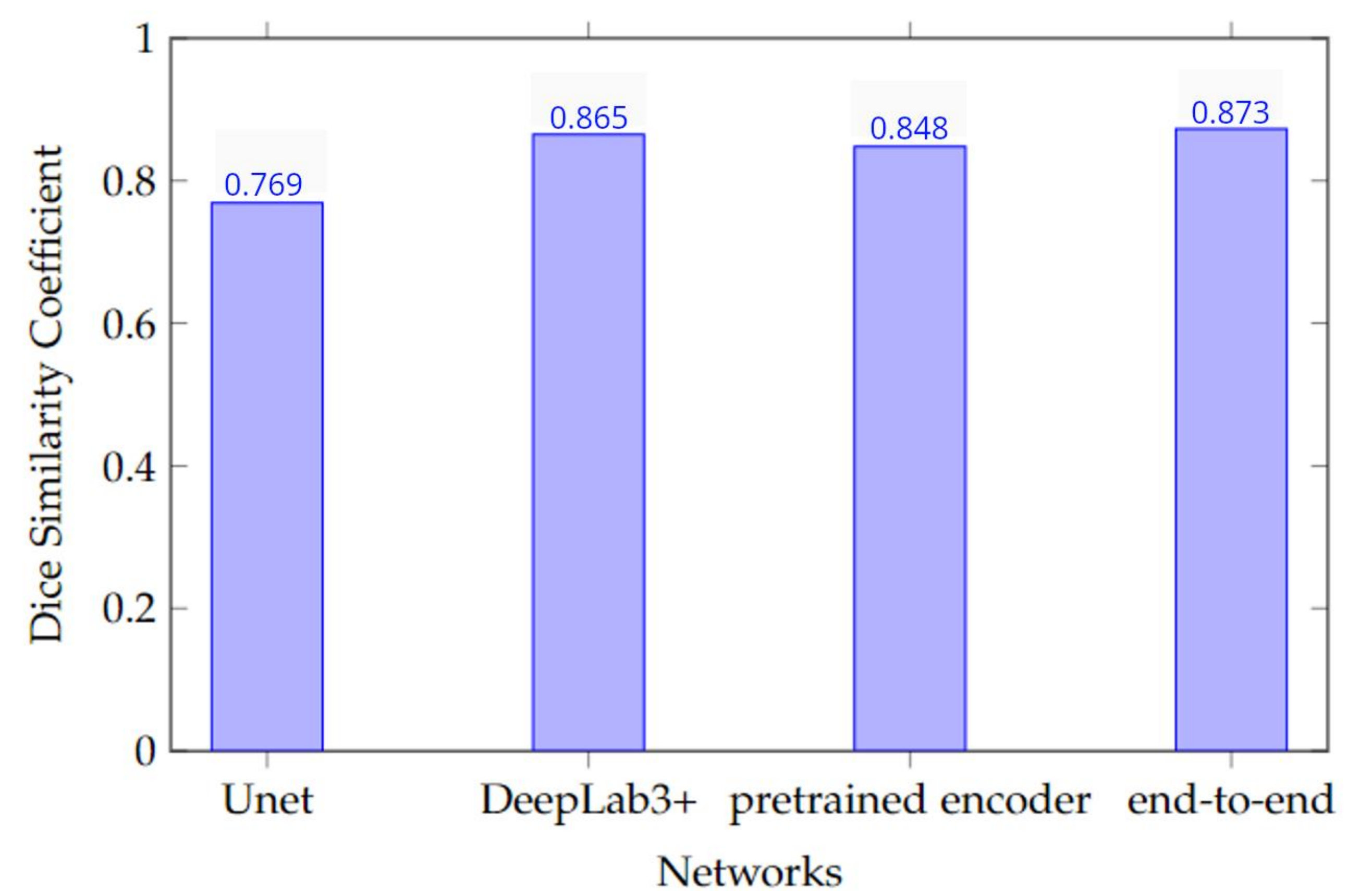
Patients	18
Vessel frames	4036
Plaque frames	3466

Our approach pre-trains the encoder on the vessel tracking data containing the frames as well as the bounding boxes for the vessel.

The complete network is trained using the segmentation data, which additionally contains the segmentation masks of the plaque.

Results

The table below shows the Dice Similarity Coefficient of our network, Unet, and DeepLab3+, validation sets on the previously mentioned dataset.



Conclusion

Initially, we created a network combining a tracking encoder and a segmentation decoder. This enabled us to focus on the vessel area through the tracking process, aiming to increase the robustness of the segmentation. Furthermore, we attempted to increase the discriminative functionality of the tracking encoder by utilizing the iterative process from the Dimp network[1]. We additionally attempted to improve the pixel-wise accuracy of the segmentation results by including the refinement modules in the decoder. Finally, we pre-trained the encoder to properly utilize its tracking functionality. The results show performance on par with the state-of-the-art network Deeplab3+.

Reference

- [1] Bhat, G., Danelljan, M., Gool, L. V., & Timofte, R. (2019). Learning discriminative model prediction for tracking. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 6182-6191).
- [2] Pinheiro, P. O., Lin, T. Y., Collobert, R., & Dollár, P. (2016). Learning to refine object segments. In *Computer Vision—ECCV 2016: 14th European conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part I 14* (pp. 75-91). Springer International Publishing.
- [3] Wang, Q., Zhang, L., Bertinetto, L., Hu, W., & Torr, P. H. (2019). Fast online object tracking and segmentation: A unifying approach. In *Proceedings of the IEEE/CVF conference on Computer Vision and Pattern Recognition* (pp. 1328-1338).
- [4] Chen, L. C., Zhu, Y., Papandreou, G., Schroff, F., & Adam, H. (2018). Encoder-decoder with atrous separable convolution for semantic image segmentation. In *Proceedings of the European conference on computer vision (ECCV)* (pp. 801-818).